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PROBLEMS ASSOCIATED WITH HOMOGENEITY TESTING IN CLIMATE VARIATION STUDIES: A CASE STUDY OF TEMPERATURE IN THE NORTHERN GREAT PLAINS, USA

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ABSTRACT

Global climate change is a controversial issue facing researchers and climatologists today. In order to obtain the most reliable results when studying climate change, the data being analysed must be as homogeneous as possible. A homogeneous time series is one in which trends and variations are caused only by effects of weather and macroclimate.

The concept of homogeneity has been addressed by some researchers, but only by testing 'average' time series such as the means and the annuals. This paper utilizes a homogeneity test developed by Alexandersson and applies it to mean monthly maximum, minimum, and mean temperature data from 22 stations in the northern Great Plains, USA. One of these, Valentine, is a first-order station and is used as the reference station. When Valentine was adjusted for a possible inhomogeneity due to its move, it was found that Valentine's adjustments had a distinct seasonal pattern.

After testing the other stations against Valentine, it was found that the position of a significant discontinuity in a station's monthly mean or annual series was not always the same in a corresponding monthly maximum and minimum series. In addition, a seasonal pattern similar to that of Valentine was found for each station's adjustment values. © 1997 the Royal Meteorological Society. Int. J. Climatol. 17: 497–510, 1997.

(No. of Figs: 6. No. of Tables: 5. No. of Refs: 4.)

KEY WORDS: homogeneity tests; temperatures; Great Plains, USA.

1. INTRODUCTION

In the study of temperature change it is necessary to ensure that the series of observations exhibits relative homogeneity. This is taken to mean that the series changes are due to climate change and not due to errors (instrumental, observer, site) or non-climatic impacts (changes in instruments, observers, screens, location, surroundings). In papers dealing with homogeneity tests of temperature series it has been common practice to investigate annual values and, if an inhomogeneity or discontinuity is identified, to adjust the series accordingly. The objective of this paper is to show that identifying a significant inhomogeneity and calculating an adjustment is by no means as straightforward as would first appear.

2. ALEXANDERSSON TEST (AT) AND THE VARIANCE

A recent study by Easterling and Peterson (1992) ranked numerous statistical tests for homogeneity of a time series. They applied the tests to mean annual temperature data from stations in continental USA. They concluded that the test developed by Alexandersson (1986) (AT) was the most effective in determining if an inhomogeneity (discontinuity) existed; it located such a discontinuity with greatest success and was the most sensitive to small discontinuities.

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The AT (1986) involves forming a sequence of differences for temperatures or ratios for precipitation, denoted as $q(i)$, between the test (candidate) station values and some type of mean value, called the reference value, from surrounding stations (or one station):

$$q(i) = \text{tsta}(i) - \text{refsta}(i), \quad (1 \leqslant i \leqslant n)$$

where tsta(i) are the test (candidate) station values and refsta(i) are the mean or reference values. This series of differences is then standardized:

$$z(i) = [q(i) - \bar{q}]/s, \quad (1 \leqslant i \leqslant n)$$

where $\bar{q}$ is the mean of $q(i)$, the difference series, and s is the sample standard deviation of the complete difference series. This new standardized series clearly has zero mean value and unit standard deviation. Two series of means $\overline{z_1}$ and $\overline{z_2}$ are then calculated, being simply the means before and after each value in the standardized series. These two series are combined into a series of values $T(v)$:

$$T(v) = [v\overline{z_1}^2 + (n - v)\overline{z_2}^2] \quad (1 \leqslant v < n)$$

where once again v varies and n is the total number of values in the standardized series. The test statistic is the maximum value in the series $T(v)$ and indicates where the largest inhomogeneity (or break) is present. The test statistic is compared with a graph of critical levels to see if the break that occurs at that point is statistically significant. If the break is significant, the series should be adjusted for the inhomogeneity.

However, there are some limitations to the AT. Easterling and Peterson (1995) noted that one limitation of the test was its weakness in finding multiple discontinuities in time series. A second limitation of the test, noted by Alexandersson himself (1986), is that significant discontinuities or inhomogeneities which occur during the first or last 5 years of the series tend to test as homogeneous because the means comparison is very biased with such small samples. In effect, this indicates that discontinuities in about the first or last five values may be missed. Also, as noted from the equations given, there is a tacit assumption that the variance remains approximately constant throughout the series. This is not necessarily true, as comparisons of variance before and after a discontinuity of data used in this study showed. In spite of these limitations it is a useful and simple test.

Table I. Meteorological stations used in this study

State	Identity number	Station name	Latitude (°N)	Longitude (°W)
Kansas	1	Colby	39·23	101·04
	2	Phillipsburg	30·45	099·19
	3	Horton	30·40	095·31
	4	Hays	38·52	099·20
	5	Colombus	37·11	094·51
Nebraska	6	Madison	41·50	097·27
	7	Fairmont	40·38	097·35
	8	Hay Springs	41·41	102·41
	9	Imperial	40·31	101·38
	10	Auburn	40·23	095·45
	11	Broken Bow	41·25	099·41
	12	Bridgeport	41·40	103·06
	13	Gothenburg	40·56	100·10
	14	Hartington	42·36	097·16
	15	Kimball	41·14	103·40
	16	Valentine	42·52	100·33
South Dakota	17	Faulkton	45·02	099·08
	18	Academy	43·30	099·04
	19	Clark	44·53	097·44
	20	Forestburg	44·02	098·04
	21	Mellette	45·09	098·30
	22	Menno	43·14	097·35

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3. DATA AND AREA

In order to study the identification of the discontinuities and their magnitude, this study considers three temperature values for each month and the annual, the mean minimum (MI), mean maximum (MA), and the mean mid-range or average of MI and MA (ME). Therefore, each station will yield 39 series, MI, MA, and ME for 12 months and the annual. It would appear at first glance that studying ME would be superfluous, however, this study will show that MI, MA, and ME can yield different results.



Figure 1. Meteorological station locations in the northern Great Plains, USA (the numbers coincide with the stations listed in Table I)

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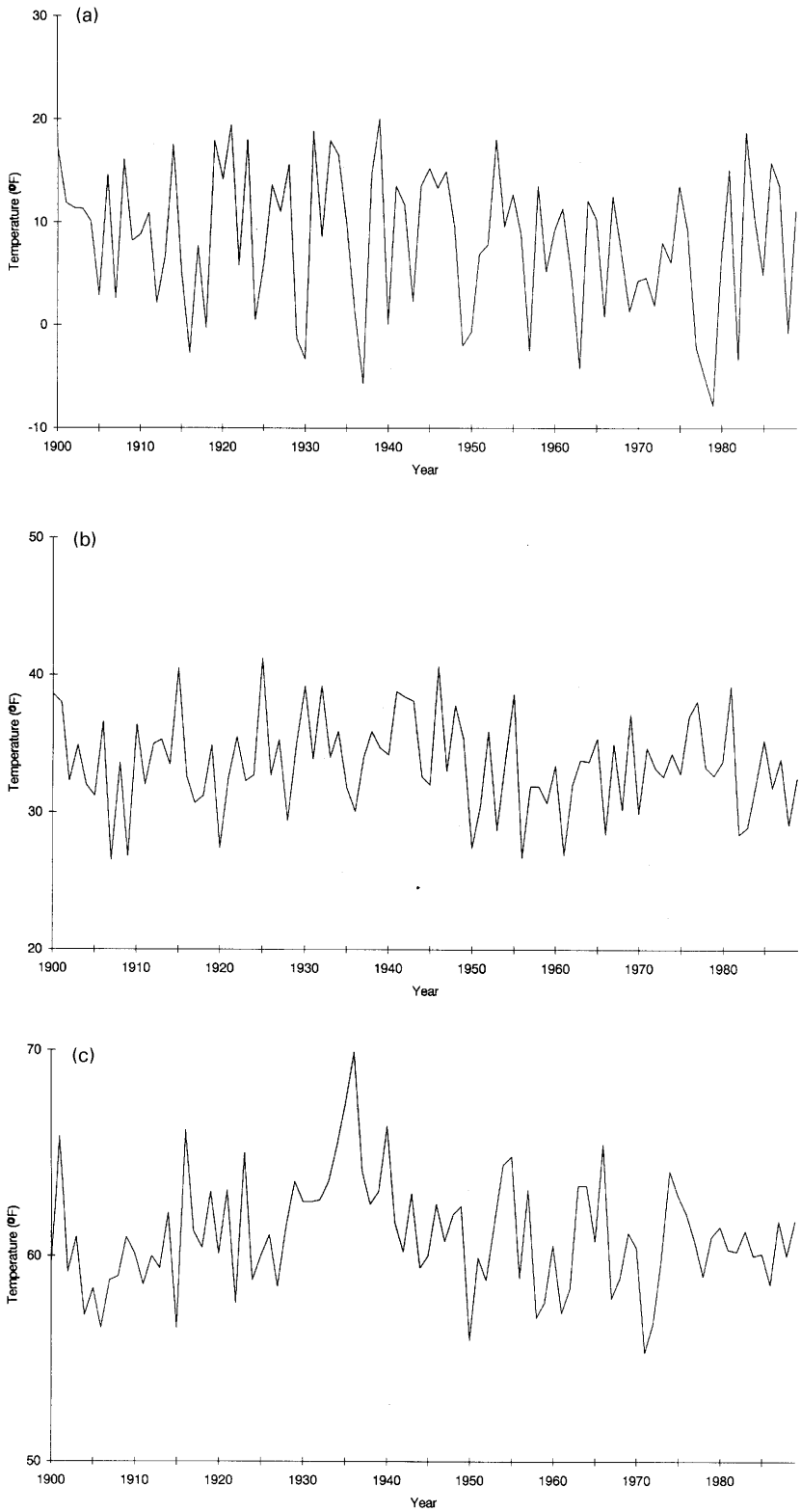


Figure 2. (a) January raw minimum (MI) temperature time series (1900–1989) for Valentine, NE. (b) As (a) but for April. (c) As (a) but for July

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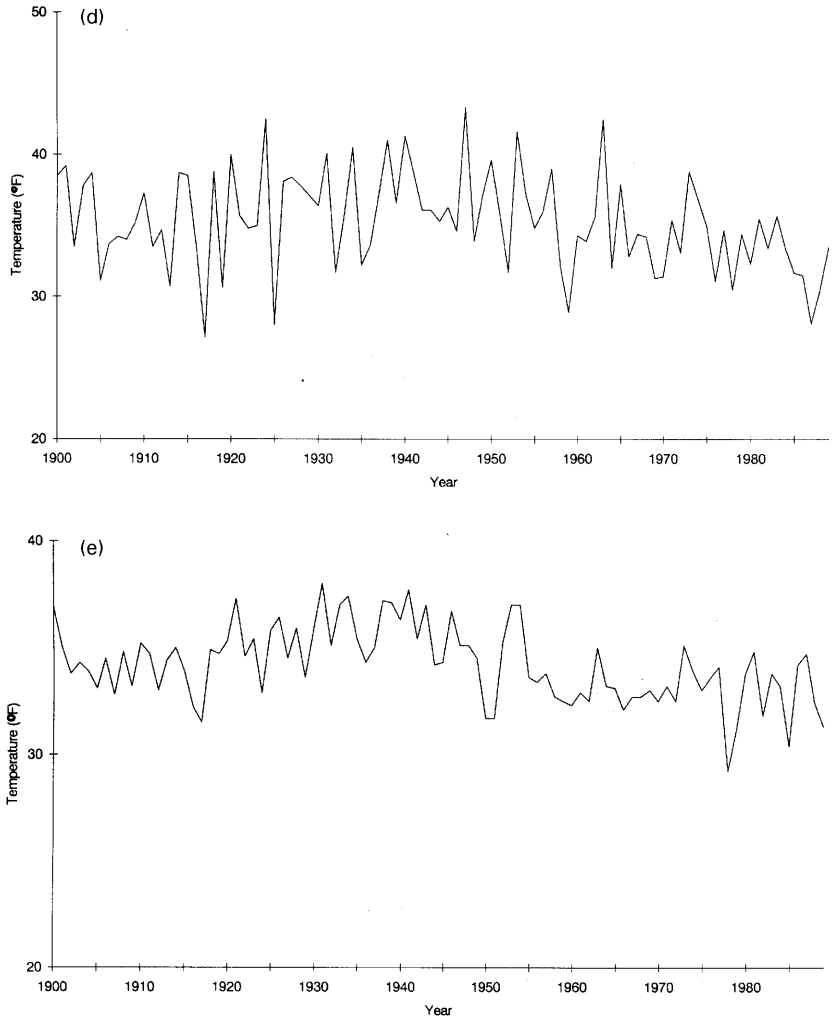


Figure 2. (d) As (a) but for October. (e) As (a) but for annual

A study of the station histories of many hundreds of climatic stations in the USA (Praner, 1985) identified 144 that had small or no relocations, few observers, and were in rural communities. Of these 22 were in the northern Great Plains and so the study focused on this area (Figure 1, Table I).

4. VALENTINE

In previous studies of homogeneity, such as Alexandersson (1986) and Easterling and Peterson (1992, 1995), the researchers combined (averaged) a large group of stations data to form their reference station data or mean values. Alexandersson (1986) took precipitation data from stations in Sweden and averaged it to form the reference station data. Easterling and Peterson (1995) used temperature data from a large group of stations near the test or candidate station to form their reference station data. The previous researchers have argued that using an average of the surrounding station data as the reference station as opposed to a single station's data is more valid when testing against the candidate or test station's data. They argue that it is risky to use one station as the reference station because its data may have inhomogeneities and when comparing it to candidate station data, it would be difficult to determine which station's data have the inhomogeneities.

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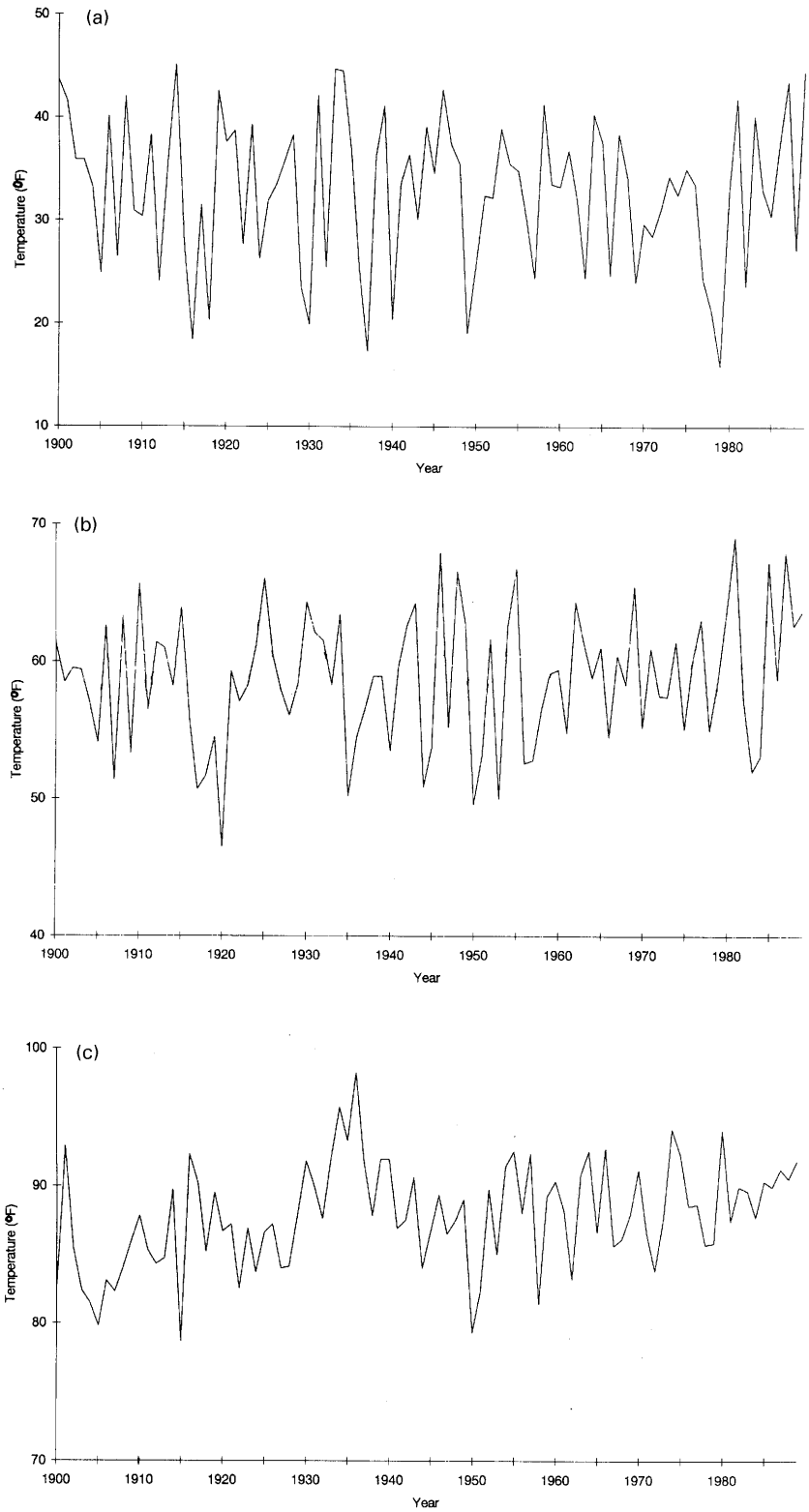


Figure 3. (a) January raw maximum (MA) temperature time series (1900–1989) for Valentine, NE. (b) As (a) but for April. (c) As (a) but for July

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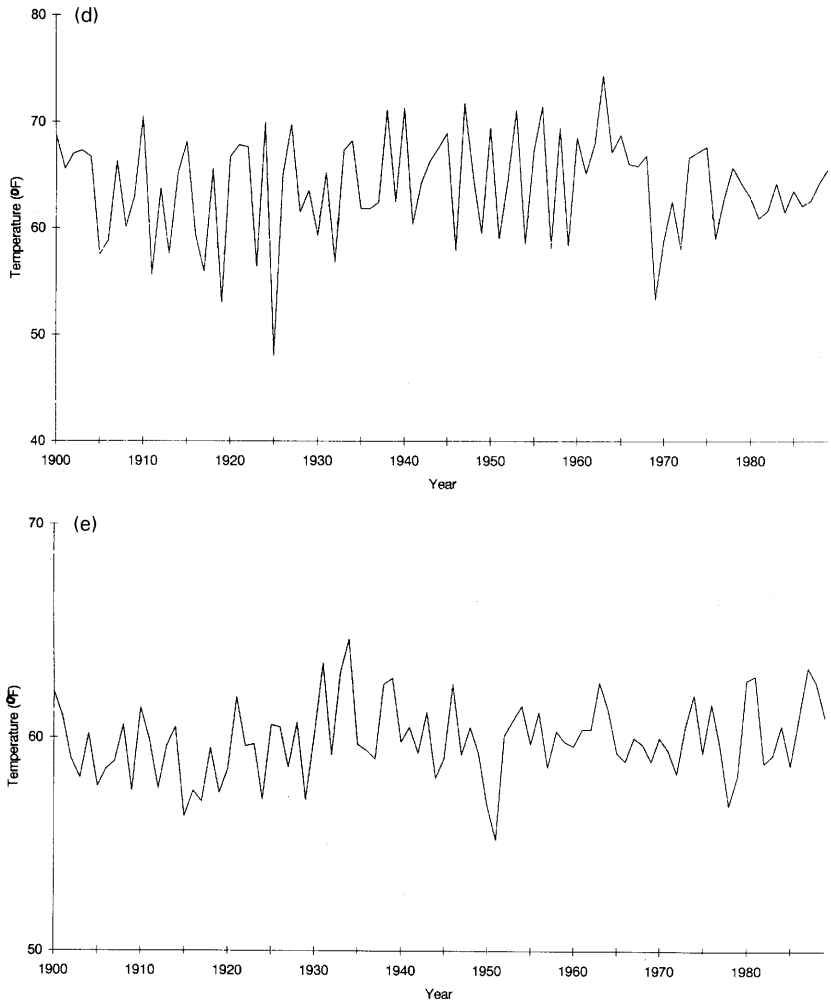


Figure 3. (d) As (a) but for October. (e) As (a) but for annual

This study takes a different approach. Using one station as a reference station is more realistic because, in many cases, a large group of good stations with complete data sets surrounding a candidate or test station is very unlikely. Many of the stations in Africa, for example, that were added from 1935 to 1985 do not necessarily have continuous, long records as they do in the USA. When studying data around the globe for inhomogeneities, researchers do not have the luxury of combining data from large amounts of stations into a reference set. It is much more realistic to choose a good, reliable station to use as a reference station, rather than hoping a large group of stations exist in the area of study.

The Valentine, NE station is the only one in this study that is classified as a first-order station by the U.S. National Weather Service. The first-order status is unique for a small, rural city with a population of less than 5000 (in 1980) and it ensures that the observers are of the highest quality. The station history for Valentine indicates that there has been only one move during the 90 years of record, namely, relocation from downtown to the nearby airport in August 1955.

Figures 2–4 show selected months and annual raw temperature time series for Valentine, NE. These monthly and annual data, along with the other months data, were used to determine if the station move had a significant effect on the homogeneity of Valentine's data. A comparison was made of the temperature means for each month and annual before (56 years) and after (34 years) the move. Table II shows the difference in means for selected months and annual before and after the move. The three temperature variables MI, MA, and ME are shown in the

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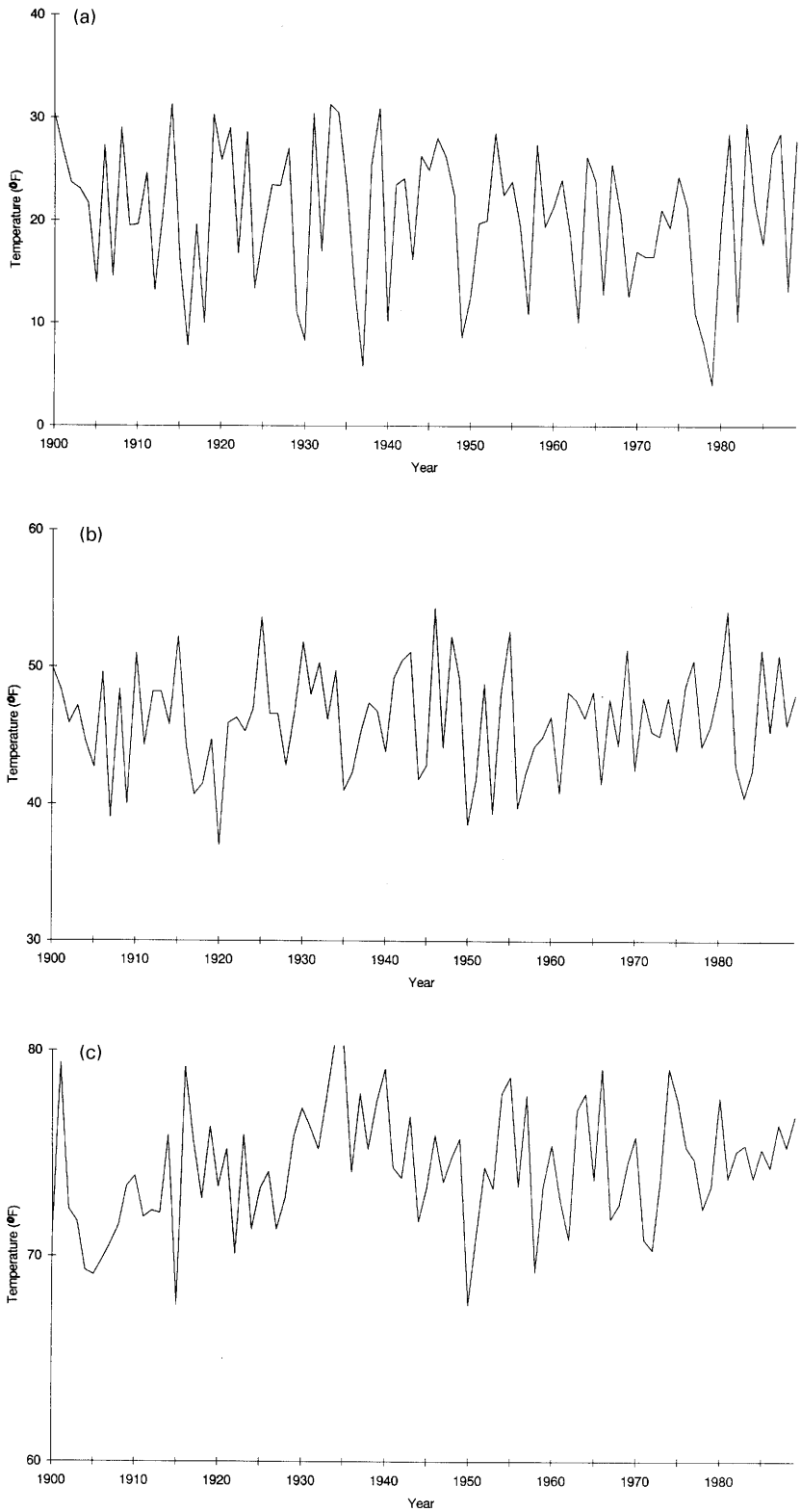


Figure 4. (a) January raw mean (ME) temperature time series (1900–1989) for Valentine, NE. (b) As (a) but for April. (c) As (a) but for July

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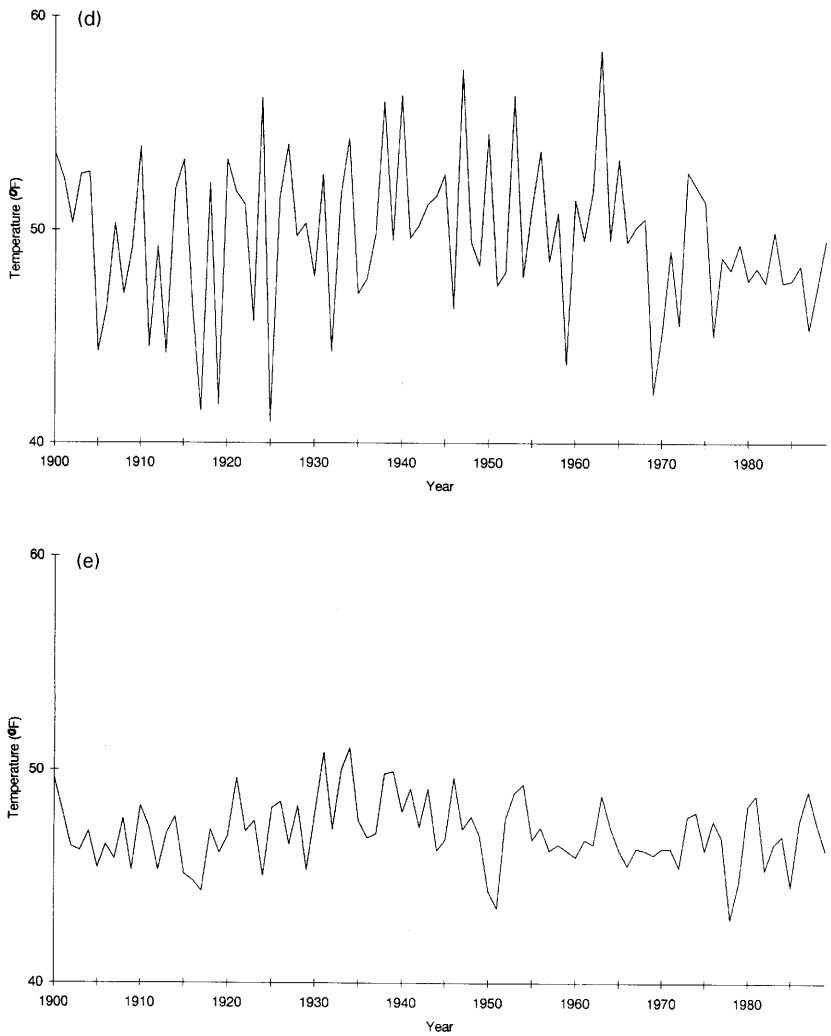


Figure 4. (d) As (a) but for October. (e) As (a) but for annual.

table as well. The variance was calculated for each monthly series before and after the move and is also included in the table. A *t*-test was performed to determine if the differences in the means were significant at the 90 per cent confidence level. An 'X is indicated if the difference in the means was significant. The results of the *t*-test show that the difference in means may be significant enough in some months to bias the results when Valentine is tested against the other stations. Therefore, the Valentine series before and after the move were adjusted so that the two have the same mean. The results of the adjustments are shown in Figure 5 (a–c). Two aspects are immediately visible, the adjustments are different among MI, MA, and ME, and there is a seasonal pattern among the adjustment values. It can be inferred from the results that making a 'correction' to monthly data from knowledge of the behaviour of the annual is not valid. Additionally, the three variables, MI, MA, and ME have independent 'corrections', although they are related because $\mathrm{MI} + \mathrm{MA} = 2\mathrm{ME}$.

5. ALEXANDERSSON TEST (AT) AND THE TEMPERATURE SERIES

With Valentine established as a reference station, the temperature series of the other 21 stations could be examined using the AT. The AT was applied to all 819 (21×39) temperature series, and 428 (52 per cent) were

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Table II. Monthly mean before (MEB) and after (MAF) Valentine, NE move and the resulting difference between them. The monthly variances before (VAB) and after (VAA) the move are also shown and whether or not the difference in means was significant at the 90 per cent critical level

		MEB	MAF	Difference	VAB	VAA	Significant
Minima	Jan	9·43	6.62	2.80	46·53	46.72	X
	April	34·13	32·58	1·55	12·54	8·97	X
	July	61·44	60·26	1·18	8·23	5·06	X
	Oct	36·17	33·89	2·28	11·79	8·63	X
	Annual	34·97	32·97	2·00	2·53	1·54	X
Maxima	Jan	33·24	32·62	0·62	56·52	50·93	
	April	58·39	59·45	− 1·06	25·01	19·84	
	July	87·22	88·77	− 1·55	17·05	9·25	X
	Oct	63·76	64·24	− 0·48	27·06	18·08	
	Annual	59·68	60·30	− 0·62	3·64	2·52	
Means	Jan	21·33	19·62	1·72	49·53	47·15	
	April	46·26	46·00	0·26	16·56	11·42	
	July	74·33	74·51	− 0·18	11·55	6·17	
	Oct	49·95	49·06	0·89	16·27	9·51	
	Annual	47·33	46·64	0·69	2·86	1·63	X

found to be homogeneous. For the non-homogeneous series, 141 applied to MI, 130 to MA, and 120 to ME. The differences among these are non-significant according to the chi-square test. The winter months (December–February) have the greatest frequency (28 per cent) of homogeneous series.

Considering the results of homogeneous series by station and temperature variable almost any outcome appears possible. Table III shows the number of monthly temperature difference series that are found to be relatively homogeneous by the AT for each temperature variable. It is clear from the table that the number of months found to be relatively homogeneous for particular stations vary among MI, MA, and ME. For example, station Broken Bow has eight months with homogeneous series in MI and MA, but when MI and MA are combined to get ME all thirteen months have homogeneous series. A similar type of increase is noted with station Menno, which goes from eight in MI and MA to ten in ME. Forestburg has a large increase, going from one in MI to five in MA, and seven in ME. However, station Mellette shows no difference among temperature variables.

Results of the AT also yielded differences in the position of significant discontinuities both among months and temperature variable for each station. Table IV shows the positions (or year preceding) of significant discontinuities for two stations Gothenburg, NE and Fairmont, NE. Individual months plus the annual and the three temperature variables for each station are shown for comparison. Differences in the positions of the significant discontinuities are evident. For instance, station Gothenburg, NE has several months plus the annual in MI where significant discontinuities occur around the same year. However, for MA and ME, the annual has a significant discontinuity between 1980 and 1981, whereas the individual inhomogeneous months have significant discontinuities that are not close to 1980–1981. On the other hand, station Fairmont, NE has a much more uniform pattern, the inhomogeneous months have significant discontinuities that are relatively close to each other and to the annual. All of the stations that were tested showed similar varying results. There were differences in position among months and when inhomogeneous months were compared with the annual. It should be noted that some of these differences may be due to the testing methods and do not necessarily represent the true nature of artificial discontinuities that may exist in the data.

The next aspect investigated was the error that may be introduced if it was assumed that the year of the discontinuity found for the annual value of a particular temperature variable was applied to all months of that series. The series then has to be compared with that series using the actual year of the discontinuity and relevant adjustment. The results are given for Fairmont in Table V, and they illustrate, once again, how the results may be misleading.

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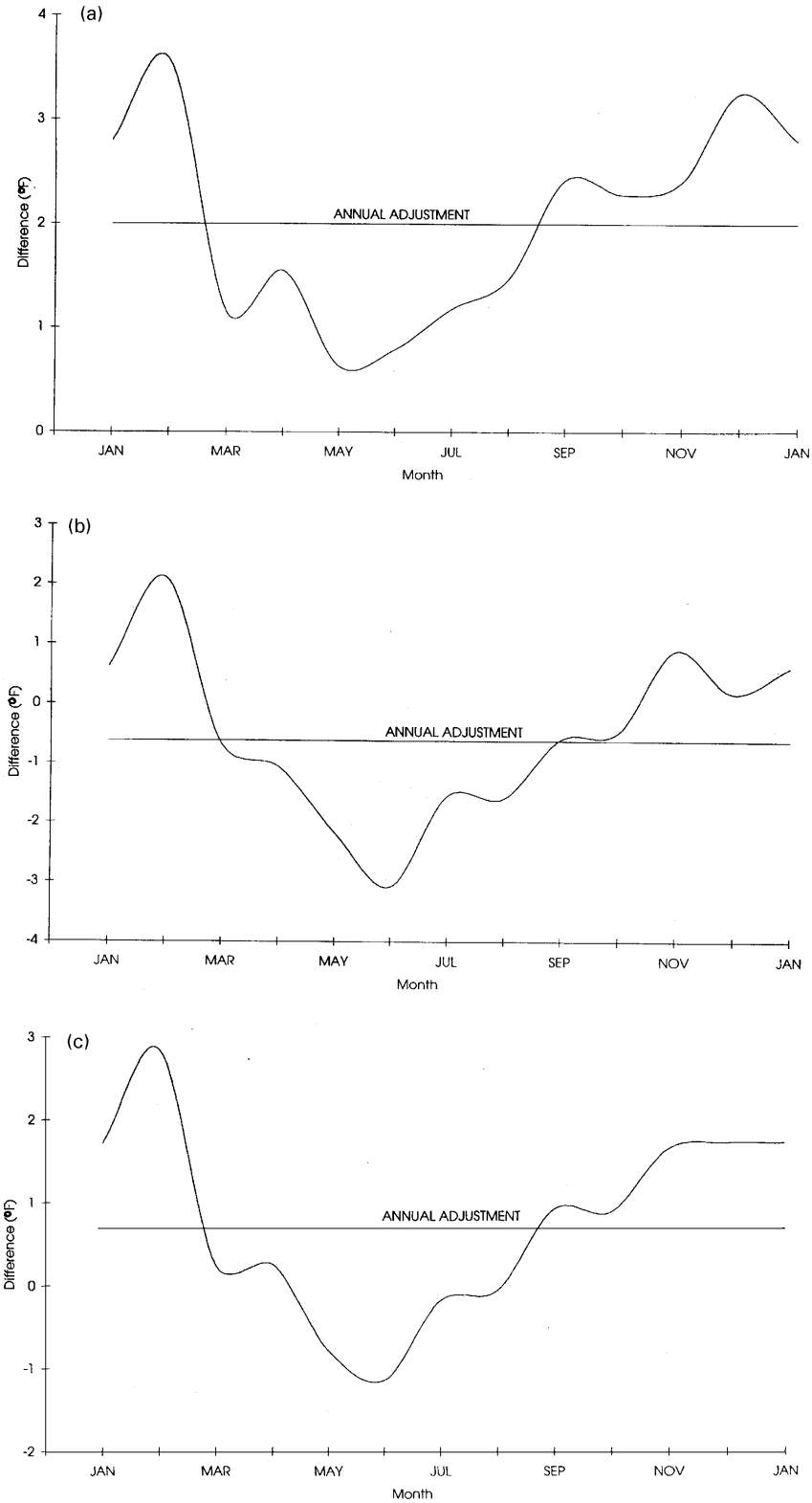


Figure 5. (a) Difference in means (or adjustment values) versus each of the months and the annual adjustment value (horizontal line) for the minimum temperatures of Valentine, NE. (b) As (a) but for maxima. (c) As (a) but for means

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Table III. The number of monthly temperature difference series for each station that are found to be relatively homogenous by the AT for each data type

	Number of monthly homogeneous series		
	Minima	Maxima	Means
Colby	3	2	3
Phillipsburg	6	7	5
Horton	8	7	8
Hays	9	9	8
Columbus	13	10	12
Madison	5	3	4
Fairmont	4	7	5
Hay Springs	7	10	7
Imperial	7	11	9
Auburn	7	8	7
Broken Bow	8	8	13
Bridgeport	7	3	4
Gothenburg	6	5	8
Hartington	5	6	6
Kimball	11	8	10
Faulkton	8	5	5
Academy	3	6	10
Clark	5	11	7
Forestburg	1	5	7
Mellette	4	4	4
Menno	8	8	10

The seasonal patterns of the adjustments are shown in Figure 6 (a–c) under the assumption that the year of the annual derived discontinuity is kept constant for all months, although that discontinuity is in a different year for MI as compared with MA and ME. There is an indication of bimodality in the three figures for which we can suggest no physical reason.

Table IV. The last two digits of the year (19XX) preceding the significant discontinuity found in each monthly difference series by the Alexandersson test for the station of Gothenburg, NE and Fairmont, NE. The dash marks (—) indicate the series that were found to be relatively homogeneous or their significant discontinuity was found during the first 5 or last 5 years

Month	Gothenburg, NE			Fairmont, NE		
	Minimum	Maximum	Means	Minimum	Maximum	Means
January	—	53	—	—	—	—
February	43	—	—	—	—	—
March	42	—	—	42	49	49
April	46	—	—	46	49	49
May	46	—	—	48	49	49
June	41	50	41	38	47	38
July	41	22	—	76	84	—
August	—	57	—	—	—	—
September	—	21	31	—	—	—
October	—	52	32	45	45	45
November	—	50	47	49	—	43
December	—	—	—	66	—	48
Annual	43	80	80	45	49	49

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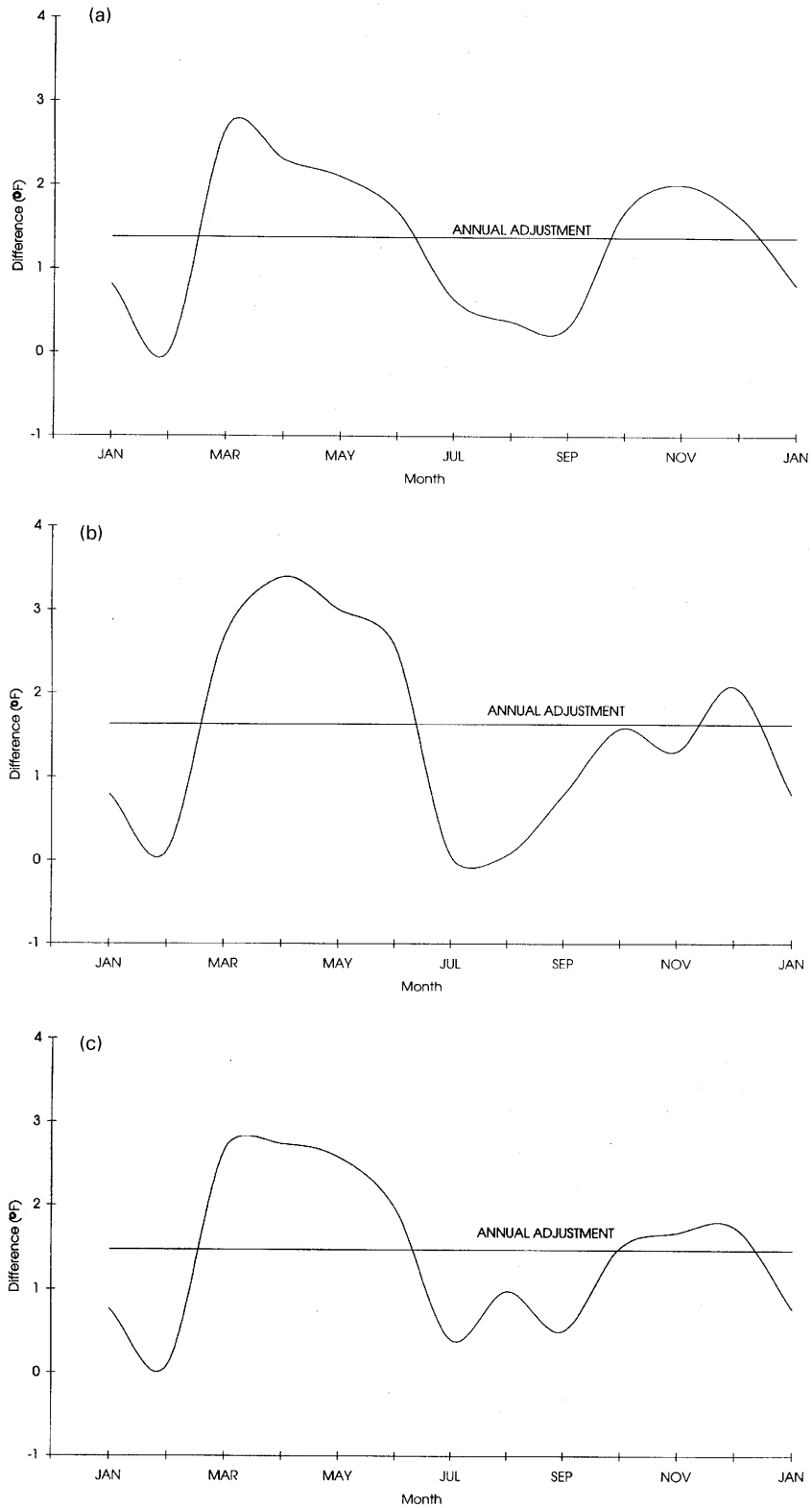


Figure 6. (a) Adjustment values versus each of the months and the annual adjustment value (horizontal line) for the minimum difference series of Fairmont, NE. (b) As (a) but for maxima. (c) As (a) but for means

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Table V. Adjustment values using the annual (AnnAdj) and the actual (ActAdj) significant discontinuity point for Fairmont, NE. The dash marks (---) indicate those series that were found to be relatively homogeneous or their significant discontinuity was found during the first 5 or last 5 years. Therefore, no adjustment amount was applied

	Minimum			Maximum			Means		
	AnnAdj	ActDis	ActAdj	AnnAdj	ActDis	ActAdj	AnnAdj	ActDis	ActAdj
Jan	0·81	—	—	0·79	—	—	0·77	—	—
Feb	0·00	—	—	0·10	—	—	0·08	—	—
Mar	2·70	42	2·76	2·68	49	2·68	2·68	49	2·68
Apr	2·30	46	2·33	3·39	49	3·39	2·74	49	2·74
May	2·10	48	2·13	3·00	49	3·00	2·58	49	2·58
June	1·68	38	2·00	2·57	47	2·56	1·96	38	2·20
July	0·63	76	1·42	0·06	—	—	0·40	—	—
Aug	0·37	—	—	0·06	—	—	0·98	—	—
Sept	0·30	—	—	0·78	—	—	0·50	—	—
Oct	1·69	45	1·69	1·58	45	1·92	1·50	45	1·81
Nov	2·01	49	2·14	1·31	—	—	1·68	43	1·70
Dec	1·64	66	2·30	2·08	—	—	1·75	48	1·81
Annual	1·38	45	1·38	1·63	49	1·69	1·47	49	1·47

6. CONCLUSIONS

The AT has been applied to a large number of temperature difference series and has produced a number of results that indicate differences among temperature variable and individual months. Although there is a large interdependence among the series, from a practical viewpoint, the differences may or may not be significant depending upon how the term 'significant' is defined. The term 'significant' will have different meanings for different investigations. This study has not attempted to explain these differences, but only expose them and raise some questions about the reliance upon mean annual data in previous homogeneity studies.

Although this study has used only one method of checking series for homogeneity, albeit the most generally accepted one, it is believed that the results bring out some serious shortcomings of the method by which inhomogeneous series are adjusted. This in turn must lead to errors in the adjusted series and cast some doubt upon the findings of some temperature change analyses.

It is noted in some of the older publications, at a time when there was the luxury of comparing readings at the old site with those at the new site over a least 1 year, that individual monthly adjustments varied greatly among themselves—and this was generally only for the means.

Of course, it is realized that the problem of multiple adjustments is physically unrealistic but perhaps we are placing too much faith upon the statistical tests used and ascribing to them an accuracy they do not warrant.

REFERENCES

Alexandersson, H. 1986. 'A homogeneity test applied to precipitation data', *J. Climatol*., **6**, 661–675.

Easterling, D. R. and Peterson, T. C. 1992. 'Techniques for detecting and adjusting for artificial discontinuities in climatological time series: a review', *Fifth International Meeting on Statistical Climatology*, 22–26 June, Toronto, Ontario, pp. J28–J32.

Easterling, D. R. and Peterson, T. C. 1995. 'A new method for detecting undocumented discontinuities in climatological time series', *Int. J. Climatol*., **15**, 369–377.

Praner, K. J. 1985. *Secular climate patterns of the north central Great Plains and the continental U.S.* , MS thesis, Texas A&M University, College Station, TX, 157 pp.

10970088, 1997, 5, Downloaded from https://rmets.onlinelibrary.wiley.com/doi/10.1002/(SICI)1097-0088(199704)17:5<497::AID-JOC140>3.0.CO;2-D by University Of Texas At, Wiley Online Library on [16/10/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License